

EXP.NO: 3. Write a Servlet program that prints "Hello, World!"

when accessed through a browser.

AIM:

To write a basic Servlet program that demonstrates the use of Java Servlets by printing

"Hello, World!" when accessed through a browser.

ALGORITHM:

1. Input:

- The user makes an HTTP request (usually by entering a URL) through a browser.

2. Servlet Program Flow:

1. Create a Servlet Class:

o Define a Java class that extends the HttpServlet class.

o Override the doGet() method, which handles HTTP GET requests from the client (browser).

2. Handle HTTP Request:

o In the doGet() method:

- Get the PrintWriter object to write the response to the browser.
- Set the response content type to text/html.
- Write the "Hello, World!" message to the response.

3. Compile the Servlet Class:

o Compile the Java Servlet file using the javac command to generate a .class file.

4. Deploy the Servlet:

o Place the compiled servlet class file in the appropriate directory of the web application (typically in the /WEB-INF/classes folder).

o Configure the servlet in the web.xml file of the web application or use

annotations.

5. Access the Servlet via Browser:

- o Start the Apache Tomcat or another servlet container.

- o Access the servlet by entering the appropriate URL (e.g.,

<http://localhost:8080/your-webapp/hello>).

Code:

```
package com.demo;
```

```
import jakarta.servlet.annotation.WebServlet;
```

```
import jakarta.servlet.http.HttpServlet;
```

```
import jakarta.servlet.http.HttpServletRequest;
```

```
import jakarta.servlet.http.HttpServletResponse;
```

```
import java.io.IOException;
```

```
import java.io.PrintWriter;
```

```
@WebServlet("/hello")
```

```
public class HelloServlet extends HttpServlet {
```

```
    private static final long serialVersionUID = 1L;
```

```
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
```

```
        throws IOException {
```

```
        response.setContentType("text/html");
```

```
        PrintWriter out = response.getWriter();
```

Sssss

```
out.println("<html>");

out.println("<head>");

out.println("<title>Hello World Animation</title>");


out.println("<style>");

out.println("body{");

out.println("background: linear-gradient(to right,#ff7e5f,#feb47b);");

out.println("font-family: Arial;");

out.println("text-align:center;");

out.println("margin-top:200px;");

out.println("}");


out.println("button{");

out.println("padding:15px 30px;");

out.println("font-size:20px;");

out.println("background:#4CAF50;");

out.println("color:white;");

out.println("border:none;");

out.println("border-radius:10px;");

out.println("cursor:pointer;");

out.println("}");


out.println("#msg{");

out.println("margin-top:30px;");

out.println("font-size:60px;");
```

Sssss

```
out.println("color:white;");
```

```
out.println("opacity:0;");
```

```
out.println("transition:opacity 1.5s ease-in;");
```

```
out.println("text-shadow:3px 3px 5px black;");
```

```
out.println("{}");
```

```
out.println("</style>");
```

```
out.println("<script>");
```

```
out.println("function showHello(){");
```

```
out.println("document.getElementById('msg').innerHTML='Hello World';");
```

```
out.println("document.getElementById('msg').style.opacity='1';");
```

```
out.println("{}");
```

```
out.println("</script>");
```

```
out.println("</head>");
```

```
out.println("<body>");
```

```
out.println("<h2>Servlet Animation Demo</h2>");
```

```
out.println("<button onclick='showHello()'>Click Me</button>");
```

```
out.println("<div id='msg'></div>");
```

```
out.println("</body>");
```

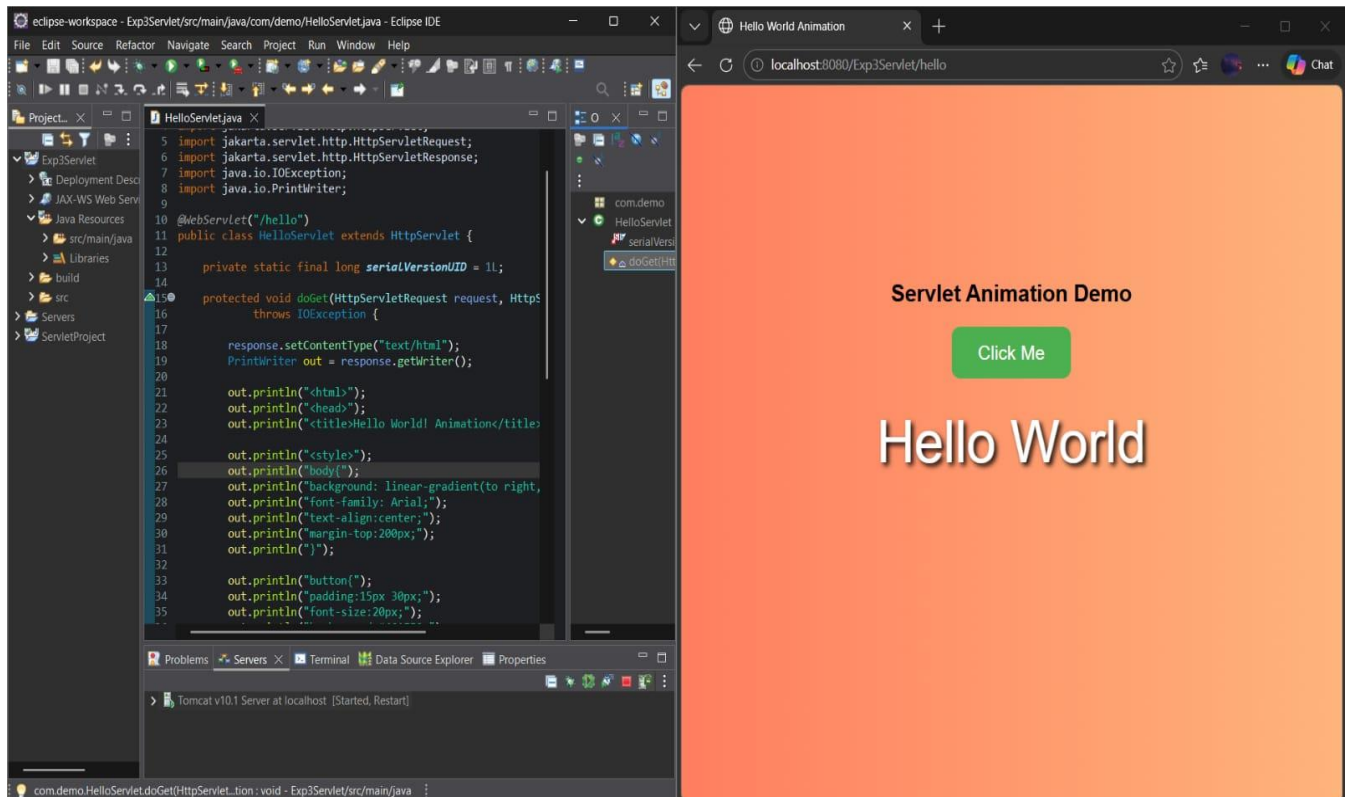
```
out.println("</html>");
```

```
}
```

```
}
```

Sssss

Output:



RESULT:

The browser will display the message "Hello, World!" returned by the Servlet.